



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
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Executive Summary

This capstone project applies data science methodologies to predict SpaceX Falcon 9 first-stage landing success, enabling competitive cost estimation for rocket launches.

Key Methodologies:

- Data collection via SpaceX REST API and Wikipedia web scraping
- Comprehensive data wrangling and exploratory data analysis
- Interactive visualizations using Folium maps and Plotly Dash dashboards
- Machine learning classification models (Logistic Regression, SVM, Decision Tree, KNN)

Key Results:

- Landing success rate improved dramatically from 0% (flights 1-9) to 70% (flights 20-49)
- KSC LC-39A achieved highest success rate at 76.9% among launch sites
- Block 5 boosters demonstrate 100% success rate with advanced technology
- Three models (Logistic Regression, SVM, KNN) achieved 83.3% prediction accuracy
- SpaceX's reusability reduces launch costs by up to 70%, from \$165M to \$67M per launch

Introduction

Project Background:

- SpaceX revolutionized aerospace economics through reusable rocket technology. Falcon 9 launches cost \$67 million compared to competitors at \$160+ million⁴. The primary cost advantage stems from first-stage reusability, which accounts for approximately 60% of total launch costs.

Business Problem:

An alternate space company needs to predict Falcon 9 first-stage landing success to determine competitive launch pricing and make informed bids against SpaceX.

Research Questions:

- What factors influence successful first-stage landings?
- Can machine learning predict landing outcomes with high accuracy?
- How do launch site, payload mass, and booster version affect success rates?
- What is the cost impact of successful vs. failed landings?"

Section 1

Methodology

Data Collection

- Data collection process involved a combination of API requests from Space X public API and web scraping data from a table in Space X's Wikipedia entry.
- The next slide will show the flowchart of data collection from API and the one after will show the flowchart of data collection from webscraping.
- **Space X API Data Columns:**
 - FlightNumber, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Flights, GridFins,
 - Reused, Legs, LandingPad, Block, ReusedCount, Serial, Longitude, Latitude
- **Wikipedia Webscrape Data Columns:**
 - Flight No., Launch site, Payload, PayloadMass, Orbit, Customer, Launch outcome, Version
Booster, Booster landing, Date, Time

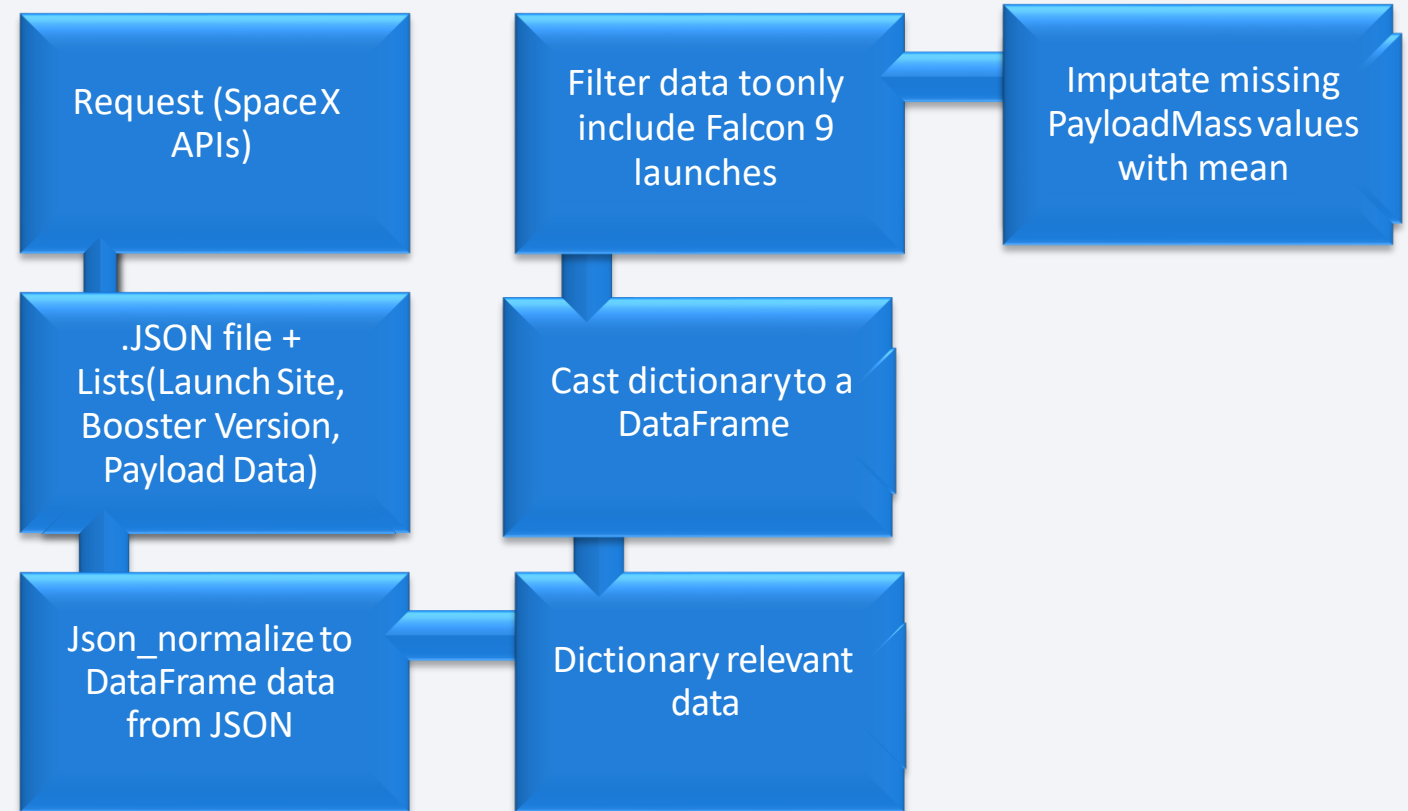
Data Collection – SpaceX API



Github URL:



[https://github.com/BlazedLith/Data-Science-IBM/tree/main/Course 10 Applied Data Science Capstone Project](https://github.com/BlazedLith/Data-Science-IBM/tree/main/Course%2010%20Applied%20Data%20Science%20Capstone%20Project)



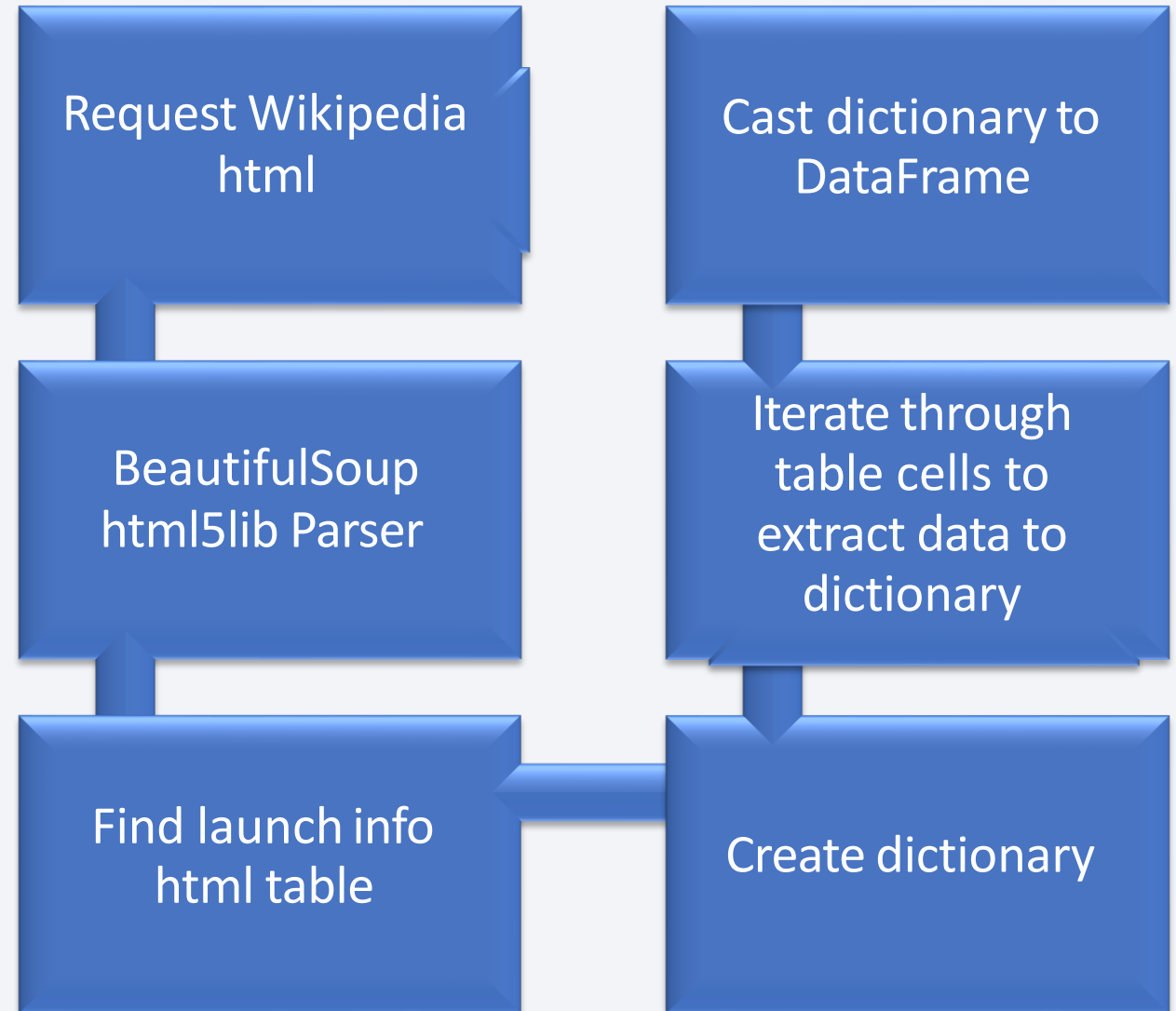
Data Collection - Scraping



Github URL:



[https://github.com/BlazedLith/Data-Science-IBM/tree/main/Course 10 Applied Data Science Capstone Project](https://github.com/BlazedLith/Data-Science-IBM/tree/main/Course%2010%20Applied%20Data%20Science%20Capstone%20Project)



Data Wrangling

- Create a training label with landing outcomes where successful = 1 & failure = 0.
- Outcome column has two components: 'Mission Outcome' 'Landing Location'
- New training label column 'class' with a value of 1 if 'Mission Outcome' is True and 0 otherwise.

Value Mapping:

- True ASDS, True RTLS, & True Ocean – set to -> 1
- None None, False ASDS, None ASDS, False Ocean, False RTLS – set to -> 0

EDA with Data Visualization

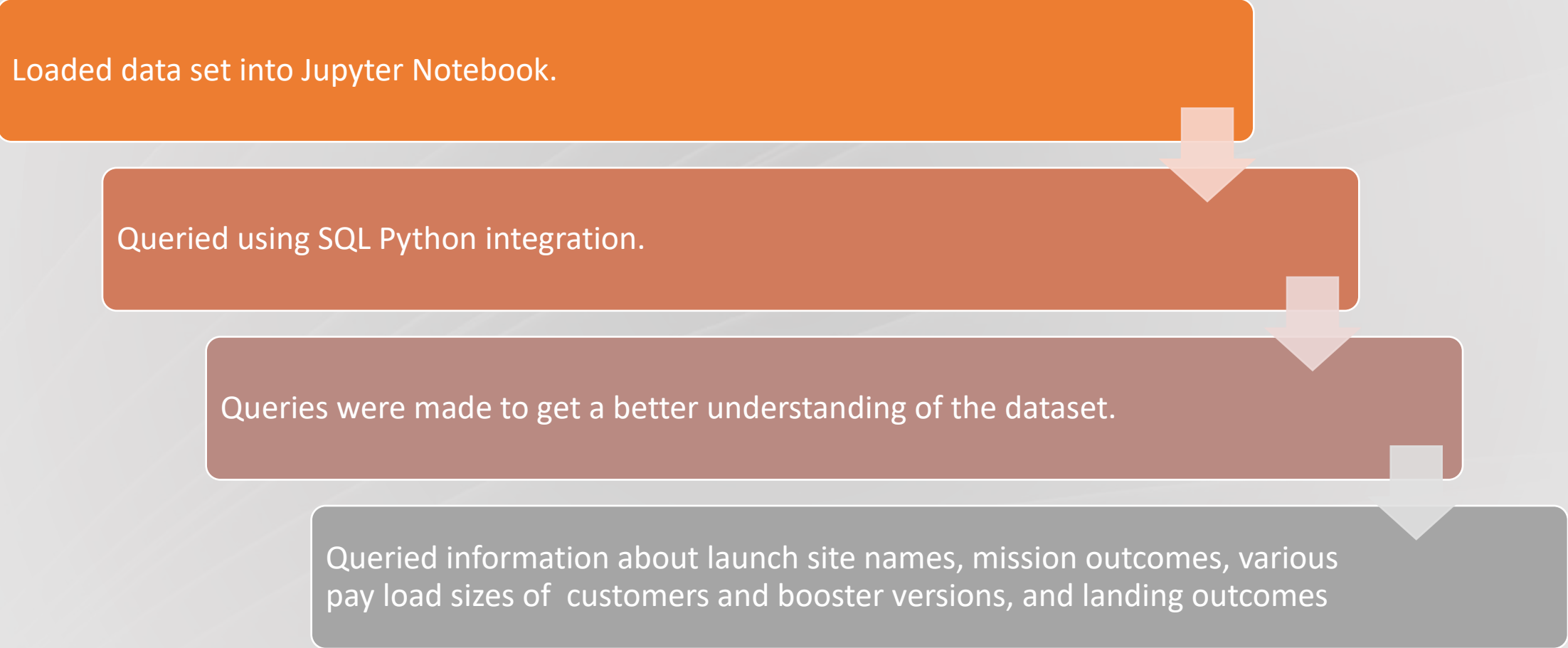
- Exploratory Data Analysis performed on variables Flight Number, Payload Mass, Launch Site, Orbit, Class and Year.

Plots Used:

- Flight Number vs. Payload Mass, Flight Number vs. Launch Site, Payload Mass vs. Launch Site, Orbit vs. Success Rate, Flight Number vs. Orbit, Payload vs Orbit, and Success Yearly Trend
- Scatter plots, line charts, and bar plots were used to compare relationships between variables to
- decide if a relationship exists so that they could be used in training the machine learning model

EDA with SQL

Loaded data set into Jupyter Notebook.



```
graph TD; A[Loaded data set into Jupyter Notebook.] --> B[Queried using SQL Python integration.]; B --> C[Queries were made to get a better understanding of the dataset.]; C --> D[Queried information about launch site names, mission outcomes, various pay load sizes of customers and booster versions, and landing outcomes];
```

Queried using SQL Python integration.

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Queried information about launch site names, mission outcomes, various pay load sizes of customers and booster versions, and landing outcomes

Build an Interactive Map with Folium

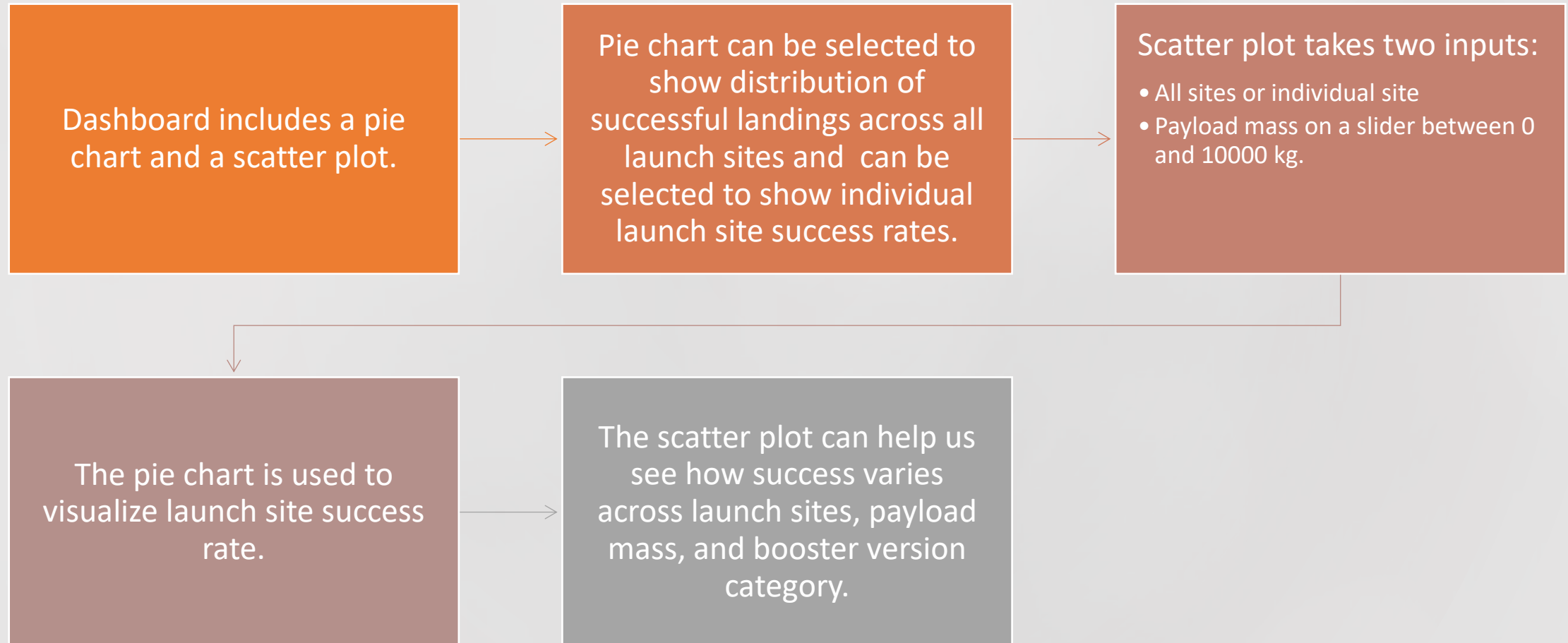


Folium maps mark Launch Sites, successful and unsuccessful landings, and a proximity example to key locations: Railway, Highway, Coast, and City.



This allows us to understand why launch sites may be located where they are. Also visualizes successful landings relative to location.

Build a Dashboard with Plotly Dash



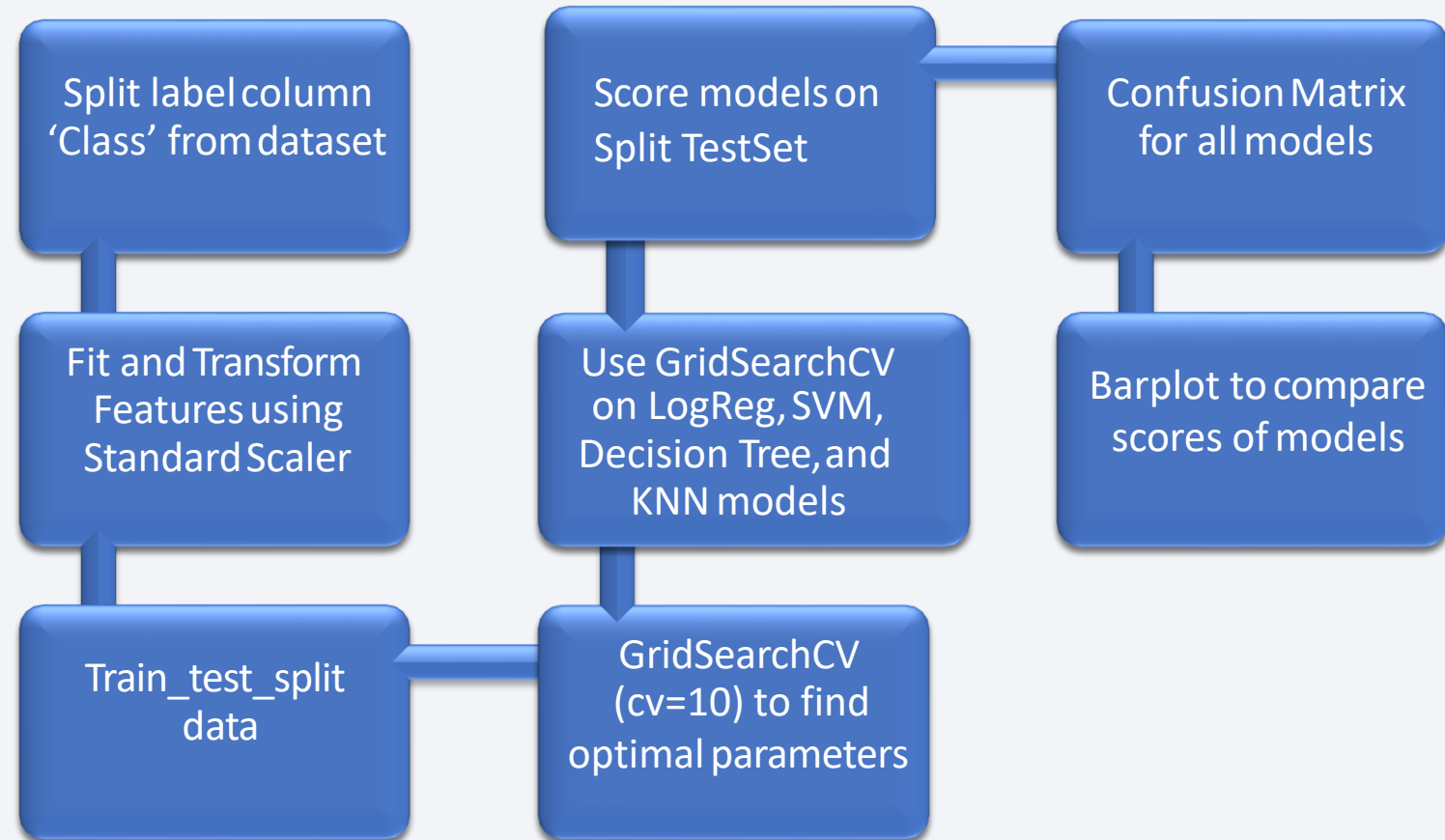
Predictive Analysis (Classification)



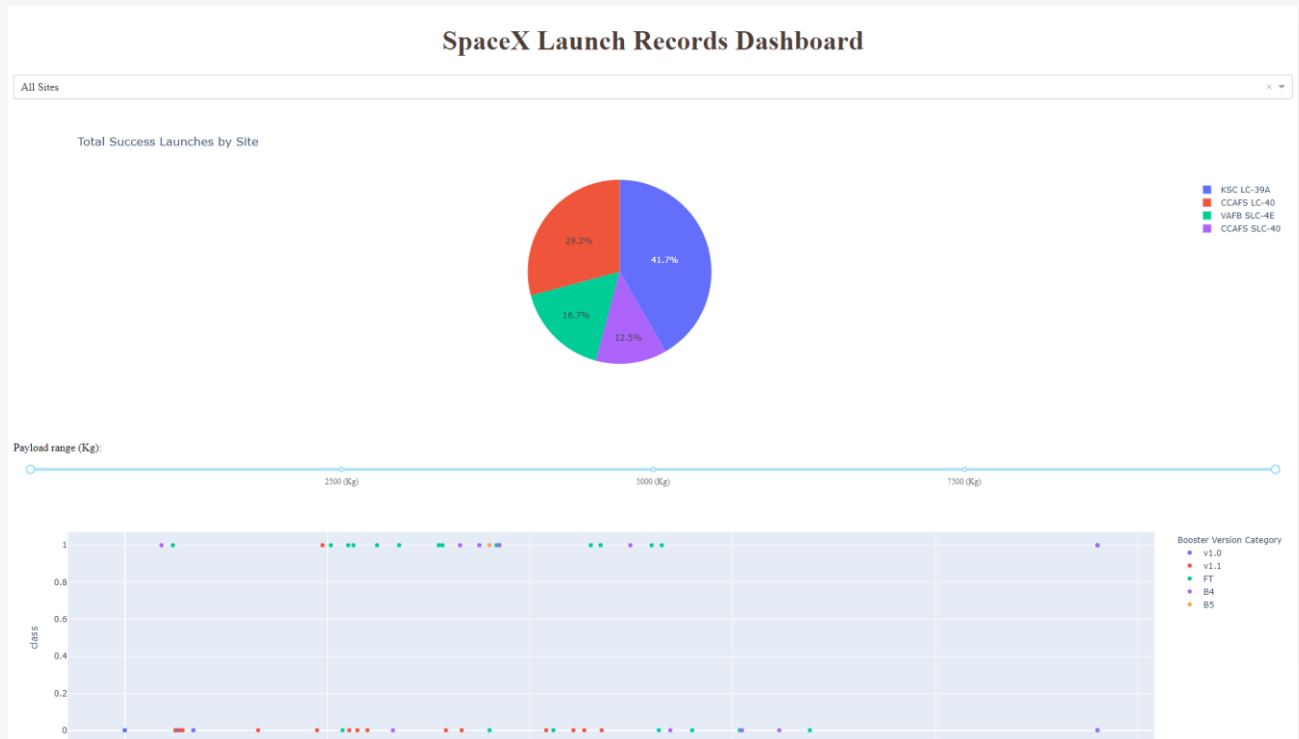
Github URL:



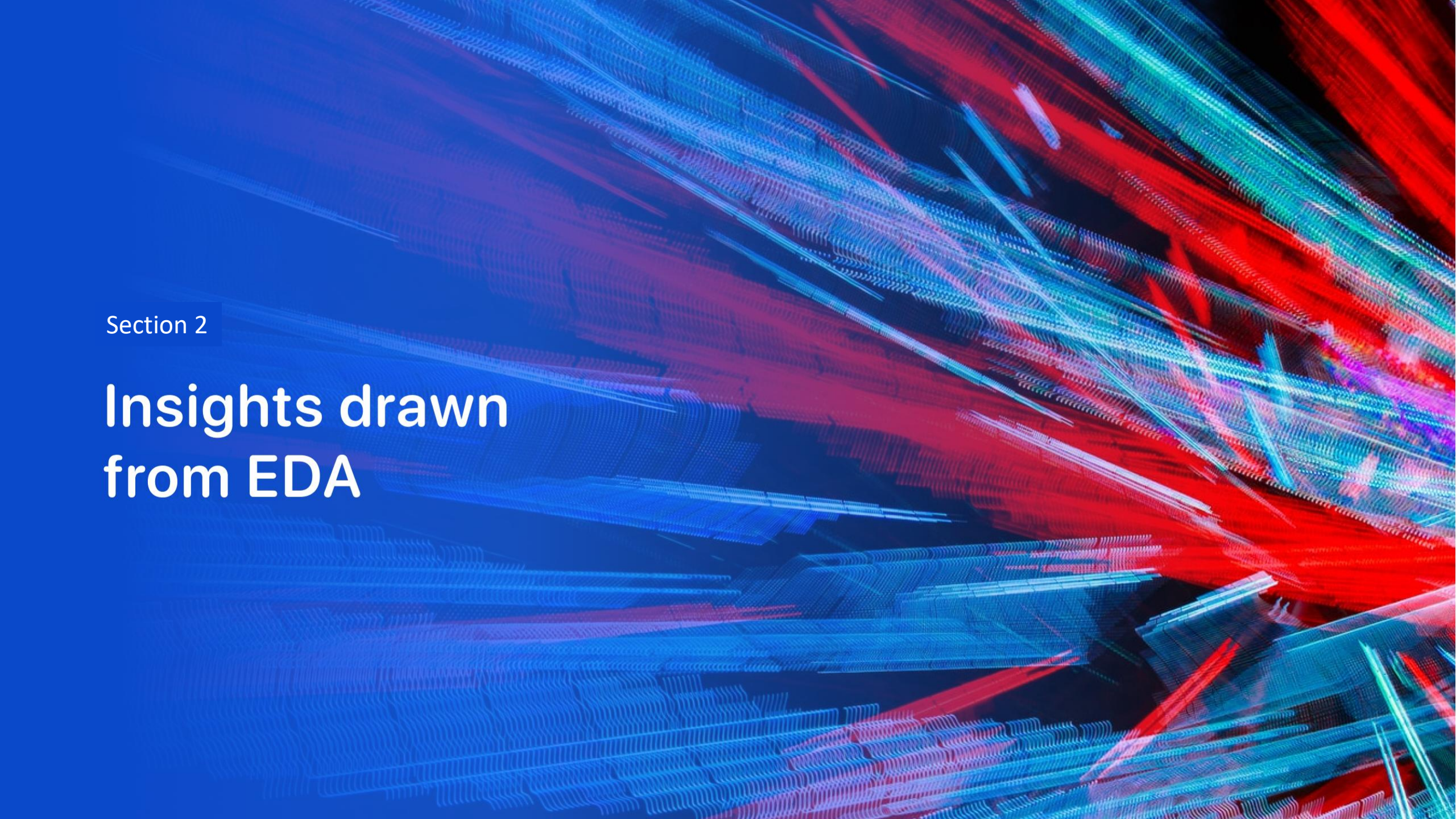
https://github.com/BlazedLith/Data-Science-IBM/tree/main/Course_10_Applied_Data_Science_Capstone_Project



Results



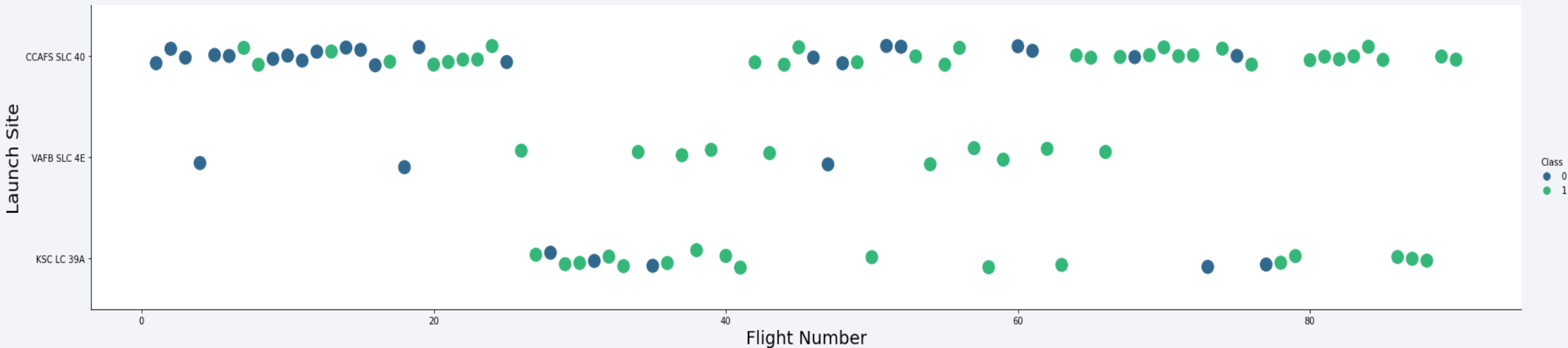
- This is a preview of the Plotly dashboard. The following slides will show the results of EDA with visualization, EDA with SQL, Interactive Map with Folium, and finally the results of our model with about 83% accuracy.

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is high-tech and digital.

Section 2

Insights drawn from EDA

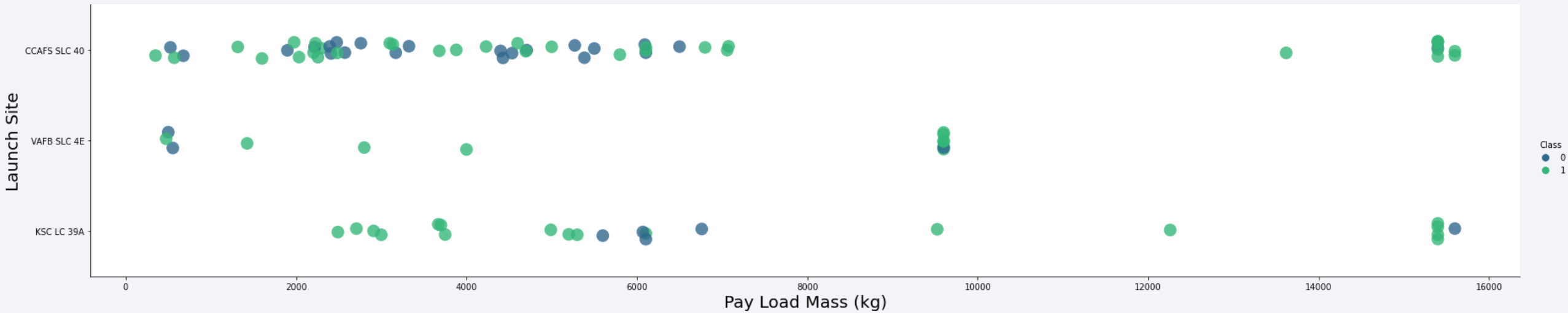
Flight Number vs. Launch Site



Graphic suggests an increase in success rate over time (indicated in Flight Number). Likely a big breakthrough around flight 20 which significantly increased success rate. CCAFS appears to be the main launch site as it has the most volume.

unch site as it has the most volume.

Payload vs. Launch Site



Payload mass appears to fall mostly between 0-6000 kg. Different launch sites also seem to use different payload mass.

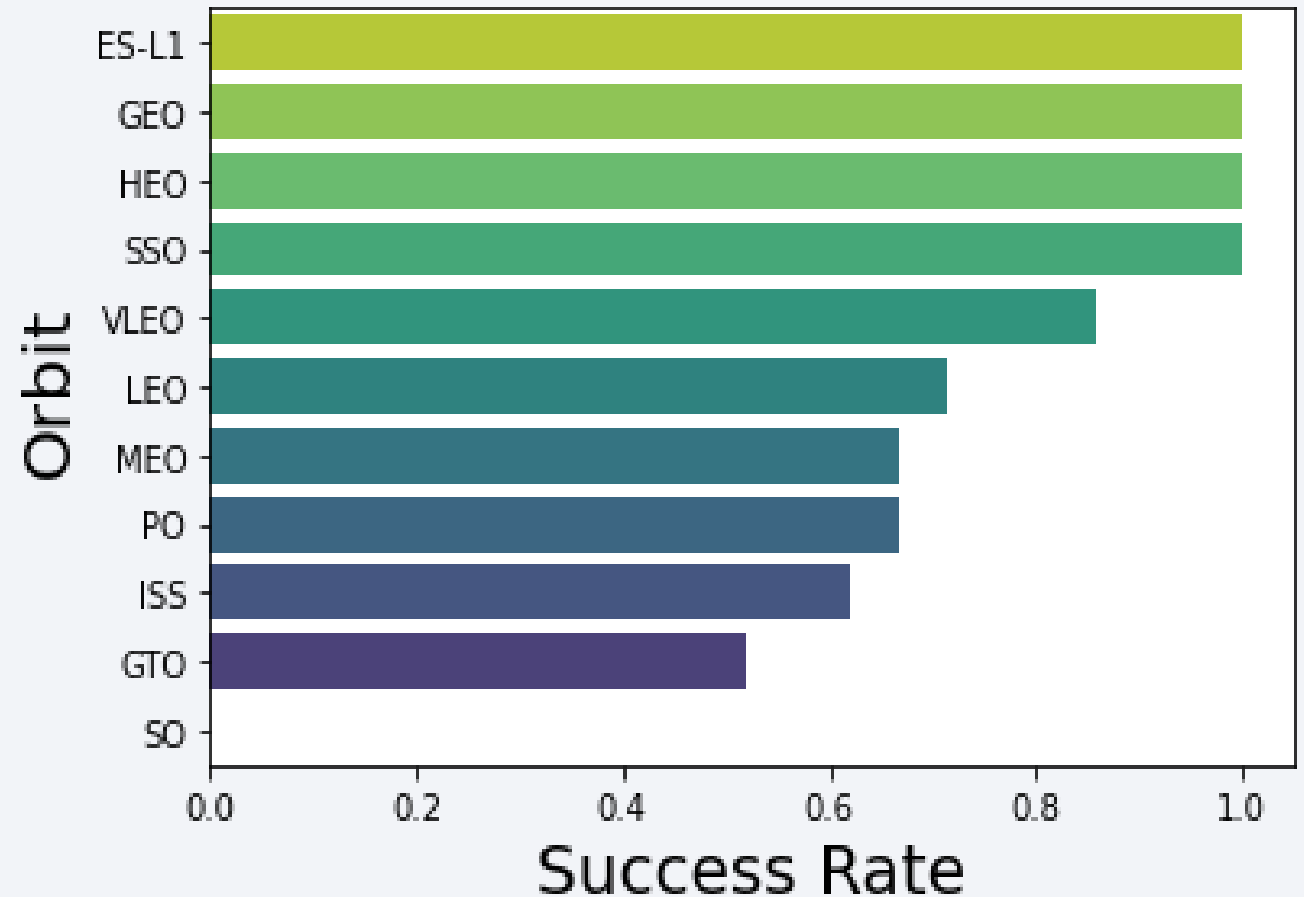
Success Rate vs. Orbit Type

ES-L1 (1), GEO (1), HEO (1) have 100% success rate (sample sizes in parenthesis)
SSO (5) has 100% success rate

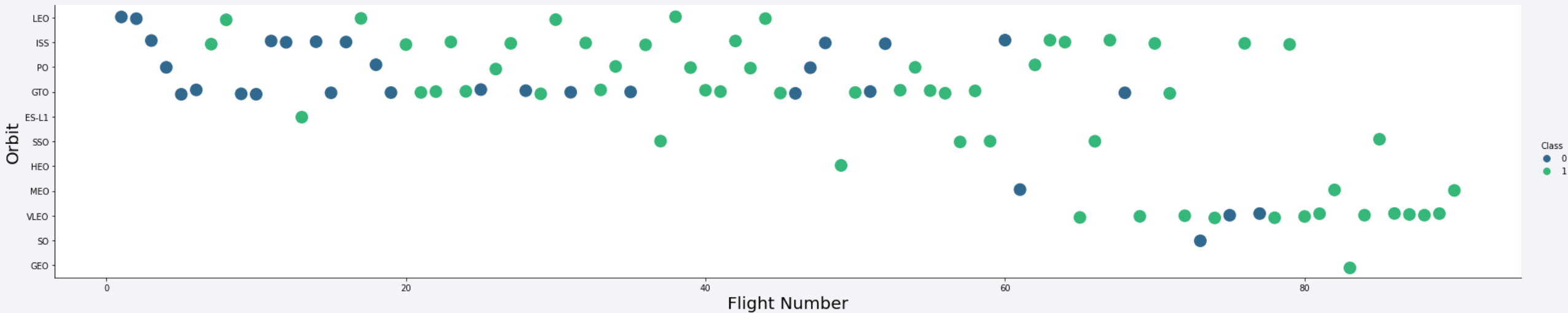
VLEO (14) has decent success rate and attempts

SO (1) has 0% success rate

GTO (27) has the around 50% success rate but largest sample



Flight Number vs. Orbit Type



Launch Orbit preferences changed over Flight Number

Launch Outcome seems to correlate with this preference

SpaceX started with LEO orbits which saw moderate success LEO and returned to VLEO in recent launches SpaceX appears to perform better in lower orbits or Sun-synchronous orbits

Payload vs. Orbit Type



Payload mass seems to correlate with orbit



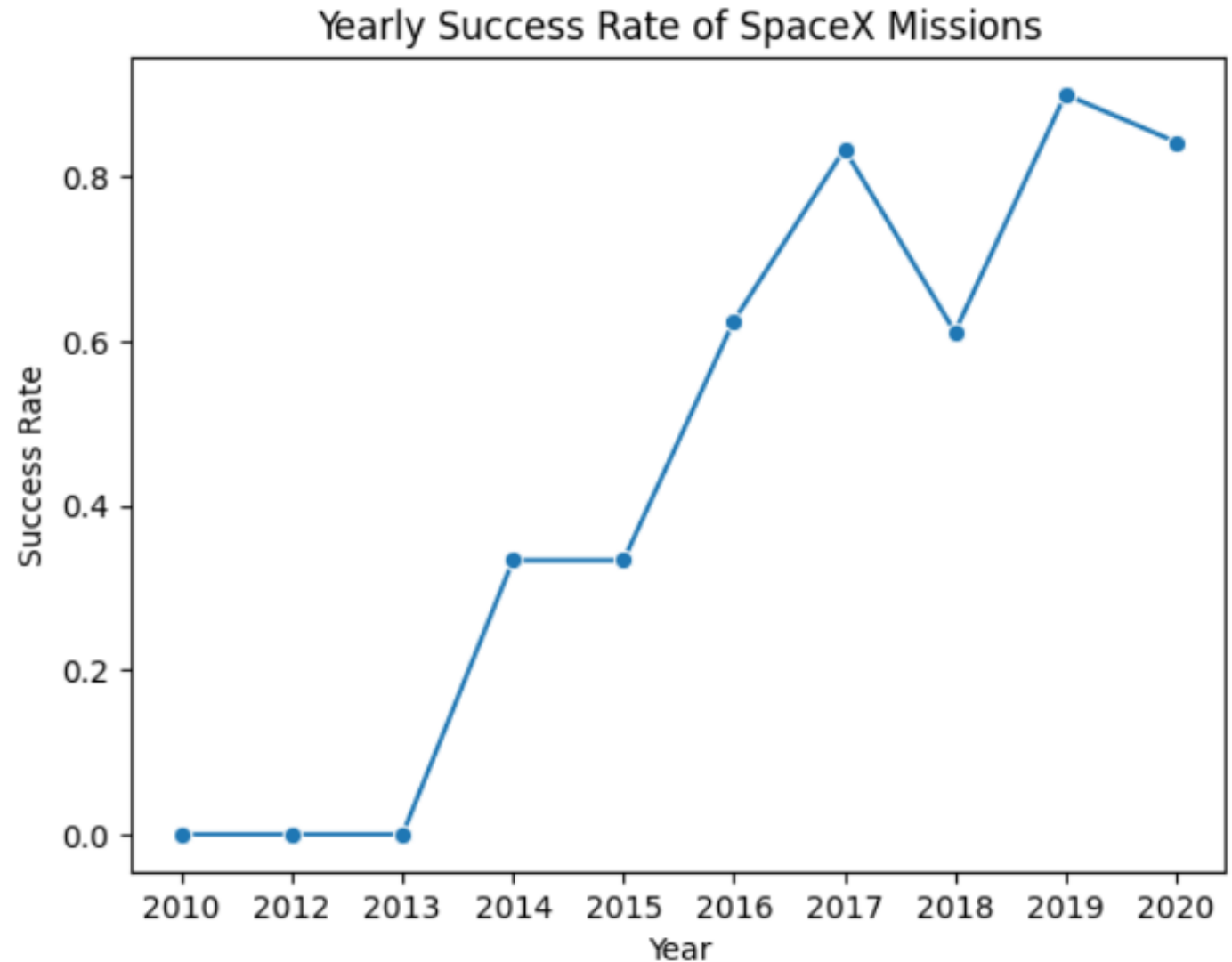
LEO and SSO seem to have relatively low payload mass



The other most successful orbit VLEO only has payload mass values in the higher end of the range

Launch Success Yearly Trend

- Success generally increases over time since 2013 with a slight dip in 2018
- Success in recent years at around 80%

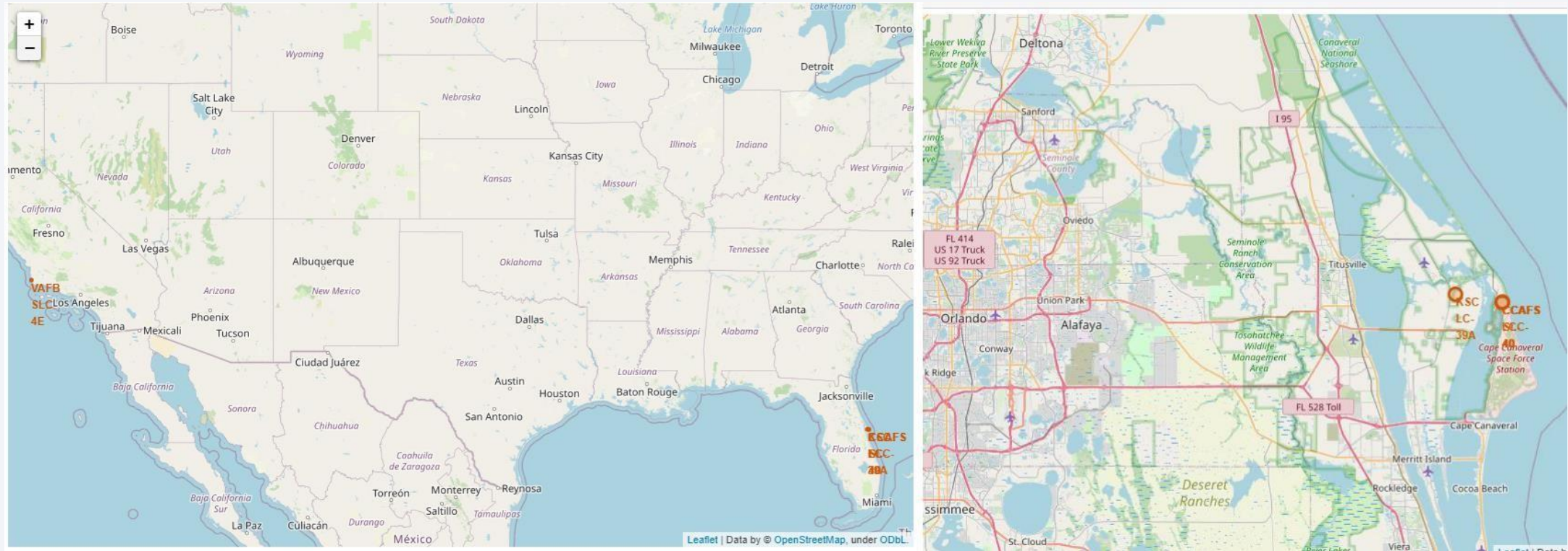


A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

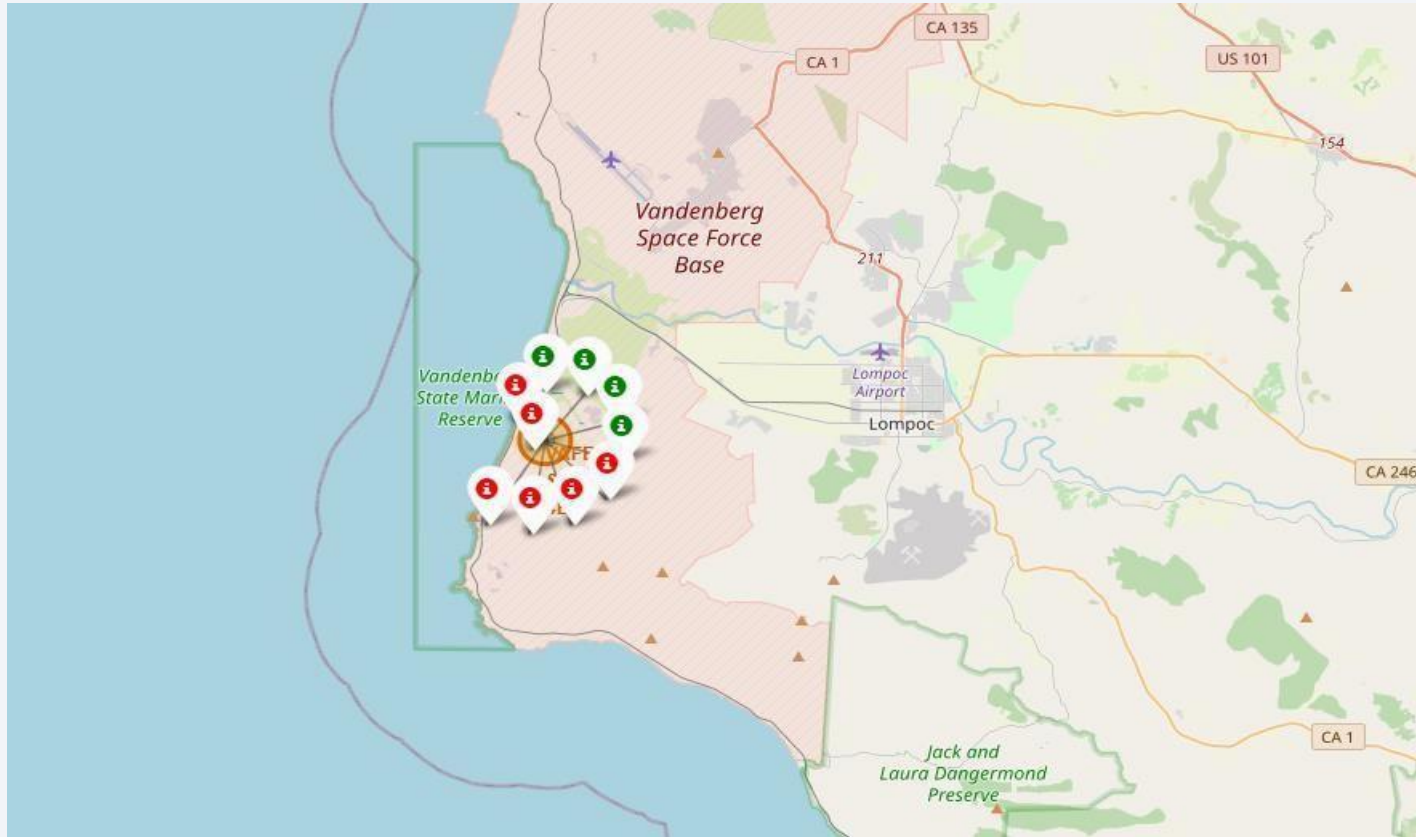
Launch Sites Proximities Analysis

Launch Site Locations



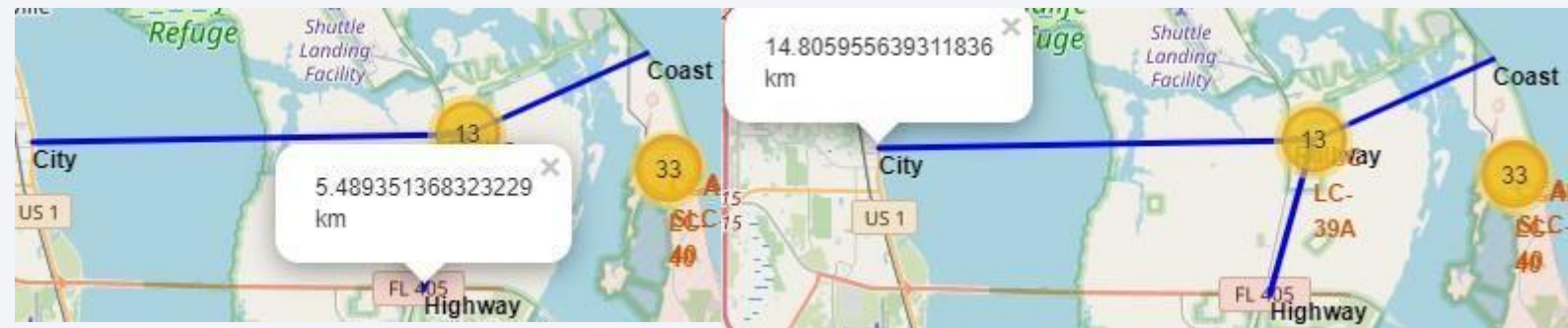
- The left map shows all launch sites relative US map. The right map shows the two Florida launch sites since they are very close to each other. All launch sites are near the ocean.

Color-Coded Launch Markers

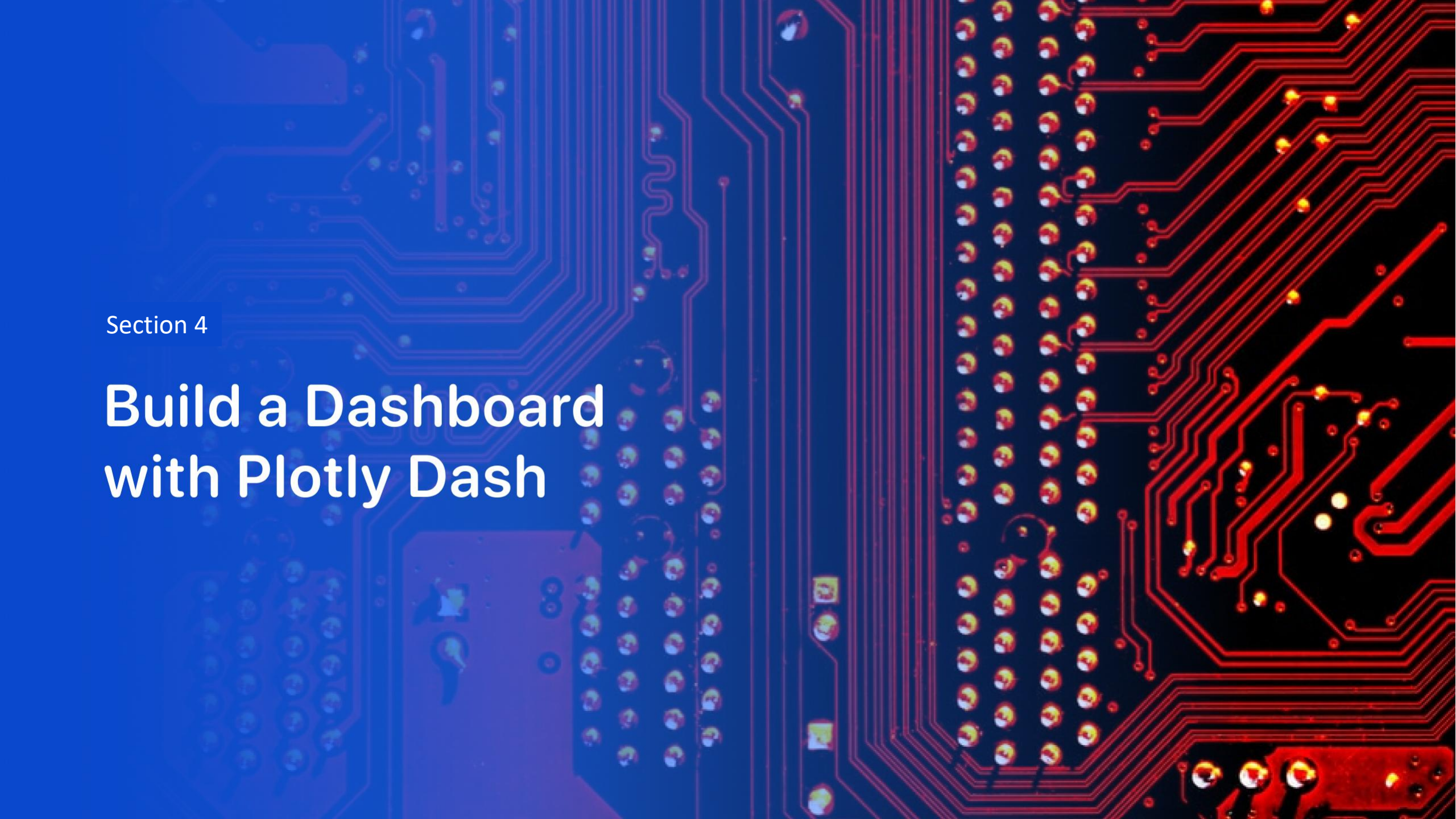


Clusters on Folium map can be clicked on to display each successful landing (green icon) and failed landing (red icon). In this example VAFB SLC-4E shows 4 successful landings and 6 failed landings.

Key Locations in Proximities



Using KSC LC-39A as an example, launch sites are very close to railways for large part and supply transportation. Launch sites are close to highways for human and supply transport. Launch sites are also close to coasts and relatively far from cities so that launch failures can land in the sea to avoid rockets falling on densely populated areas.

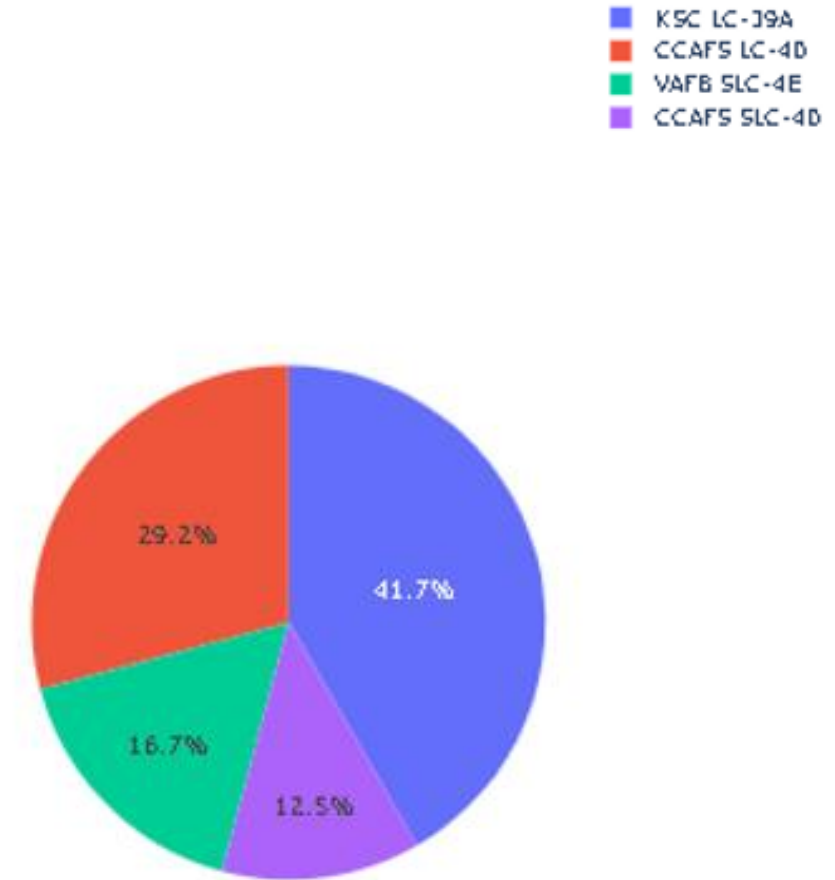


Section 4

Build a Dashboard with Plotly Dash

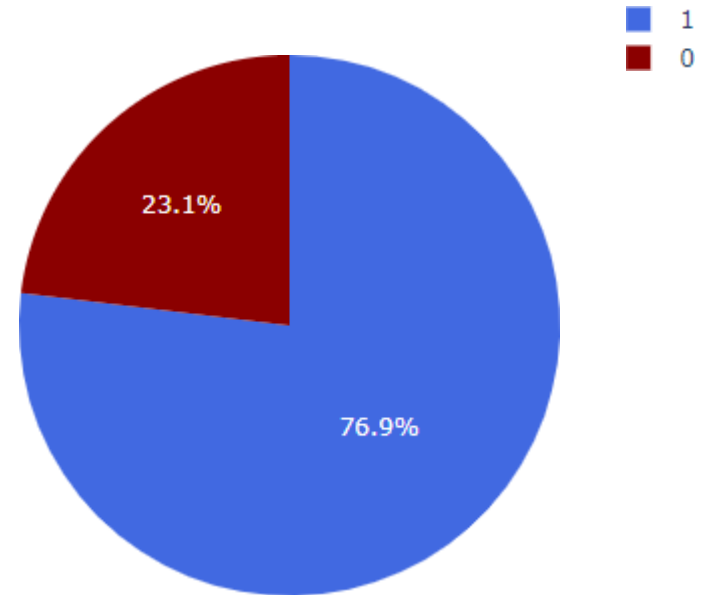
Successful Launches Across Sites

- This is the distribution of successful landings across all launch sites. CCAFS LC-40 is the old name of CCAFS SLC-40 so CCAFS and KSC have the same amount of successful landings, but a majority of the successful landings were performed before the name change. VAFB has the smallest share of successful landings. This may be due to smaller sample and increase in difficulty of launching in the west coast.



Successful Launches Across Sites

- KSC LC-39A has the highest success rate with 10 successful landings and 3 failed landings.

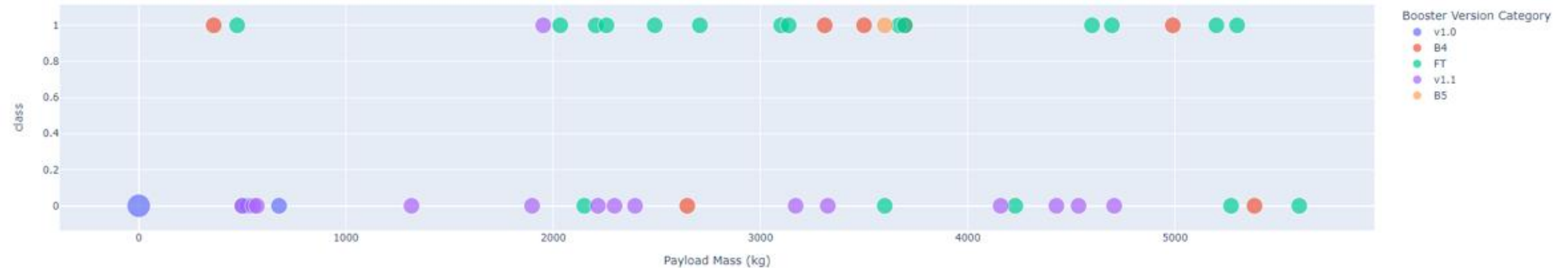


KSC LC-39A Success Rate (blue=success)

Payload range (Kg):



Payload Mass vs. Success vs. Booster Version Category



Payload Mass vs Booster Version vs Success

Plotly dashboard has a Payload range selector. However, this is set from 0-10000 instead of the max Payload of 15600. Class indicates 1 for successful landing and 0 for failure. Scatter plot also accounts for booster version category in color and number of launches in point size. In this particular range of 0-6000, interestingly there are two failed landings with payloads of zero kg.



Section 5

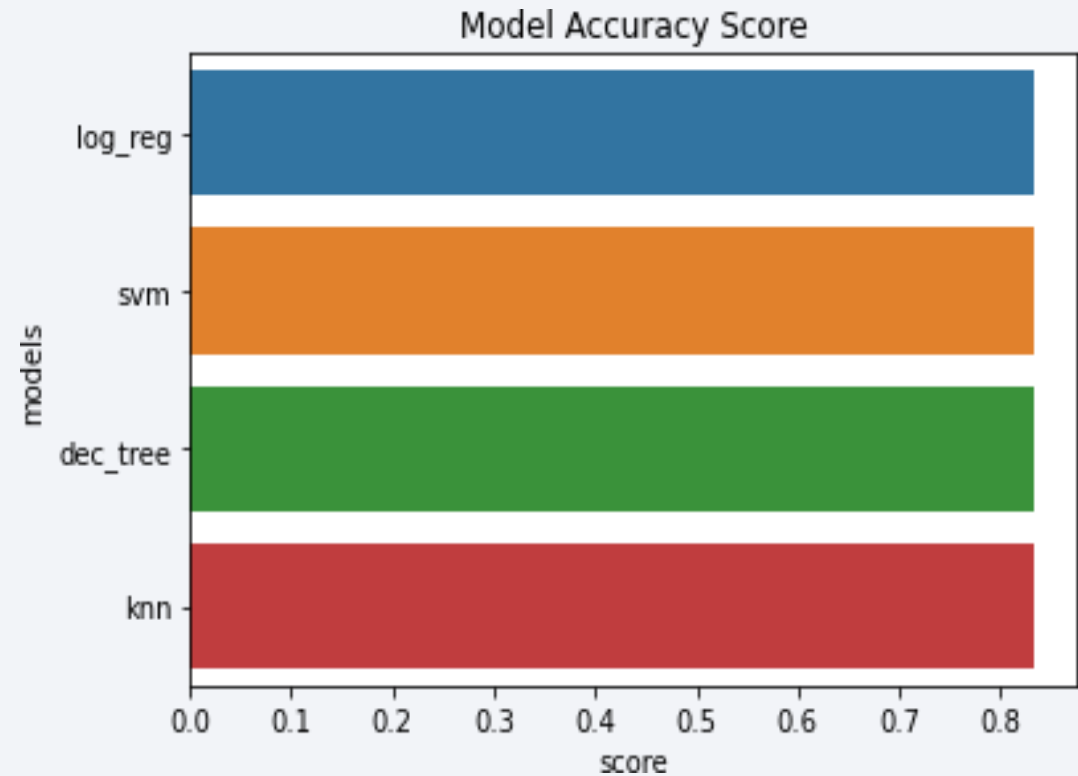
Predictive Analysis (Classification)

Classification Accuracy

All models had virtually the same accuracy on the test set at 83.33% accuracy. It should be noted that test size is small at only sample size of 18.

This can cause large variance in accuracy results, such as those in Decision Tree Classifier model in repeated runs.

We likely need more data to determine the best model.



Confusion Matrix

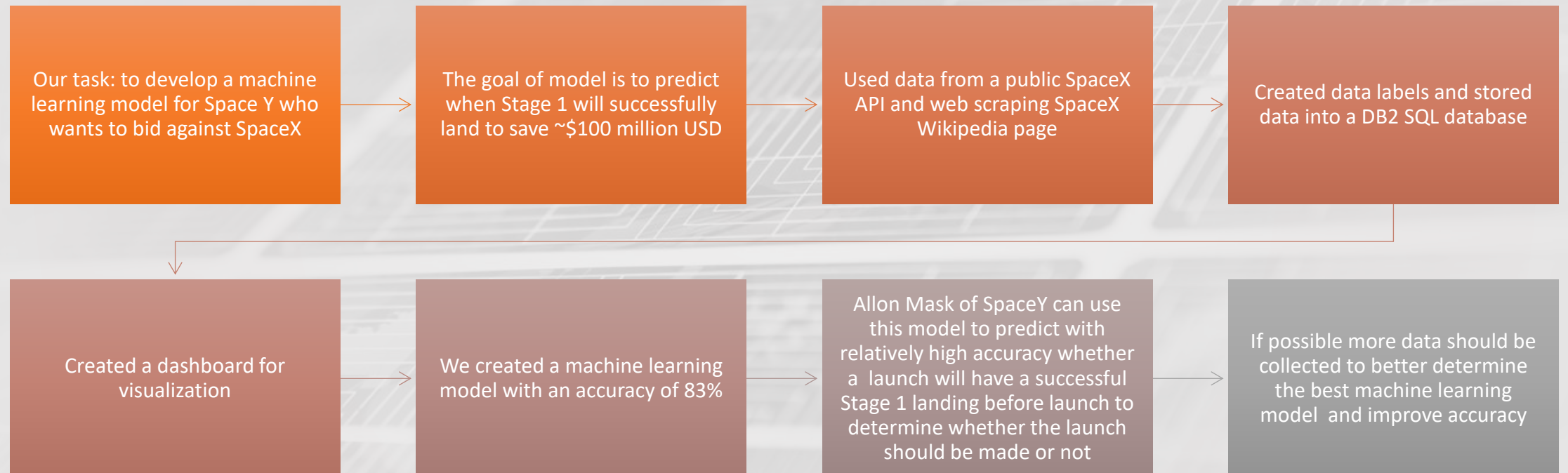
Since all models performed the same for the test set, the confusion matrix is the same across all models. The models predicted 12 successful landings when the true label was successful landing.

The models predicted 3 unsuccessful landings when the true label was unsuccessful landing.

The models predicted 3 successful landings when the true label was unsuccessful landings (false positives). Our models over predict successful landings.



Conclusions



Appendix

GitHub repository url:

- [https://github.com/BlazedLith/Data-Science-IBM/tree/main/Course 10 Applied Data Science Capstone Project](https://github.com/BlazedLith/Data-Science-IBM/tree/main/Course%2010%20Applied%20Data%20Science%20Capstone%20Project)

Instructors:

- Rav Ahuja, Alex Aklson, Aije Egwaikhide, Svetlana Levitan, Romeo Kienzler, Polong Lin, Joseph Santarcangelo, Azim Hirjani, Hima Vasudevan, Saishruthi Swaminathan, Saeed Aghabozorgi, Yan Luo

Special Thanks to All Instructors:

- <https://www.coursera.org/professional-certificates/ibm-data-science?#instructors>

Thank you!

